

Workshop – 8

Ishwor Chalise

Installing ELK With Ubuntu and Nginx With Filebeat

```
ishwor@OHH:~/bigdata$ curl -fsSL https://artifacts.elastic.co/GPG-KEY-elasticsearch
-----BEGIN PGP PUBLIC KEY BLOCK-----

mQENBFi3HsoBCADXDtbNjnxBPqB1vDNtCsqhe49vFYsZN9IOZsZXgp7aHjh6CJBD
A+bGFOwyhbd7at35jQjWAw103cFYsKAmFy+Ar3LHCMkV3oZspJACTIgCrwnkic/9
CULiQe324qv0bU2QRtP4FL0zWcfb/S8UYzWXWIFuJqMvE9MaRY1bwUBvzoqavLGZ
j3SF1SP0+TB5QrHkrQHBsmX+Jda6d4Ylt8/t6CvMwgQNlrlzIO9WT+YN6zS+sqHd
1YK/ay5qhoLnhp9G/HxhcSVCKLq8SStj1ZZ1S9juBPoXV1ZWNBxFNGwOh/NYGLdD
2kmBf3YgCqeLzHahsAEpvAm8TBa7Q9W21C8vABEBAG0RUVsYXN0aWNzZWYyY2gg
KEVsYXN0aWNzZWYyY2ggU2lnbmLuZyBLZXkpIDxkZXZfb3BzQGVSyXN0aWNzZWYy
Y2gub3JnPokBTgQTAAQgAOAIbAwIXgBYhBEYJWsyFSFgsGiaZqdJ9ZmzYjkK0BQJk
9vrZBQsJCAcDBRUKCQgLBRYCAwEAAh4FAAoJENJ9ZmzYjkK00hoH+wYXZKgVb3Wv
4AA/+T1IAf7edwgaJR58bEyqds6/4v6uZBneUaqahUqMXgLFRX5dBSrAS7bvE/jx
+BBQx+rpFGxSwvFegRevElzAGVtpgkFQX0RpRcKSmksucSBxikR/dPn9XdJSEVa8
vPcs11V+2E5tq3LEP14zJL4MkJKQF0VJL5UUmKLS7U2F/IB5aXry9UwDMTnwNntX
kl2iDaViYf4M6C6xTS24uLwND2St0Jvjt+xGEwbdBVvp+UZ/kG6IGkYM5eWGPuok/
DHvjUdwTfy09b5xGbqn5F3JUFOWB/nOSFXHM8rsHRT/67gHcIl8YFqSQXpIkk9D3
dCY+KieW0ue5AQ0EUjceygeIAIOSVJc3DFuf3LsmUfGpUmnCqoUm76Eqm8xynFEG
ZpczTChkwARRtckcfa/sv376j+jk0c0Q71Uv3MnMLPGF+w3bpu8fLiPeW/cntf1
8uZ6DxJvHa/oaZZ6VPjwUGSeVydipTzFTYsce08Dx13gpS6nHZ9Gsnfr/kcH9/11
Ca73HBtmGVikOI1mZKMbAN08cewY/i7fPxShu7B0Rb3jxVNGUuiRcfRiao0gWx0U
ZGpvuHpl7l0FX2cbsHFAp9WsjYebSohb/Y0K4NkyFhL82MfbcsEwsXPhRTFgJWw
s4vpuFg/kFFlnw0NPNVP1jNjLNCsMBMEpP1A7k6MRpylNnUAEEAAAYkBNgQYAQgA
IAIbDBYhBEYJWsyFSFgsGiaZqdJ9ZmzYjkK0BQJk9vsHAAoJENJ9ZmzYjkK0hwsh
/ArKtn12HM3+41zYo9q04rTri7+IYTjSB/JDT0usZgZLd/HCP1xQo4S12Eur3Rtx
USMwK1LEebZsjwDT9yVceYekrBEqUVyRMSVYj+UeZK2s4LbXm9b4jxXVtaivmkMA
jtznndrD7kmm8ak+UsZplf6p6uZS9TZ9hjwoMmw5oMaS6TZkLT4KYGWeyzHJSUBX
YikY6vssDQu4SJ07m1f4Hz81J39Q0cHln5I5HTK8Rh/VUFCxNnGg9360g55wWpiF
eUTeMyoXp0tffiUhi0tBRysmSYC0D4Fd5yJn03n1pwnVVVsM7RAC22rc5j/Dw8dR
GIHikRcYWeXTYW7veewK5S=
=ftS0
-----END PGP PUBLIC KEY BLOCK-----
```

```
ishwor@OHH:~/bigdata$ curl -fsSL https://artifacts.elastic.co/GPG-KEY-elasticsearch |sudo gpg --dear
mor -o /usr/share/keyrings/elastic.gpg
[sudo] password for ishwor:
File '/usr/share/keyrings/elastic.gpg' exists. Overwrite? (y/N) y
ishwor@OHH:~/bigdata$ echo "deb [signed-by=/usr/share/keyrings/elastic.gpg] https://artifacts.elastic.
co/packages/7.x/apt stable main" | sudo tee -a /etc/apt/sources.list.d/elastic-7.x.list
deb [signed-by=/usr/share/keyrings/elastic.gpg] https://artifacts.elastic.co/packages/7.x/apt stable m
ain
```

```
ishwor@OHH:~/bigdata$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu jammy InRelease
Get:2 https://artifacts.elastic.co/packages/7.x/apt stable InRelease [13.7 kB]
Hit:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://security.ubuntu.com/ubuntu jammy-security InRelease
Get:5 https://artifacts.elastic.co/packages/7.x/apt stable/main amd64 Packages [145 kB]
Hit:6 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:7 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu jammy InRelease
Fetched 158 kB in 2s (91.6 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
55 packages can be upgraded. Run 'apt list --upgradable' to see them.
ishwor@OHH:~/bigdata$ sudo apt install elasticsearch
```

```
ishwor@OHH:~/bigdata$ sudo apt install elasticsearch
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  http libnl-genl-3-200
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  elasticsearch
0 upgraded, 1 newly installed, 0 to remove and 55 not upgraded.
Need to get 325 MB of archives.
After this operation, 542 MB of additional disk space will be used.
Get:1 https://artifacts.elastic.co/packages/7.x/apt/stable/main amd64 elasticsearch amd64 7.17.29 [325
  MB]
Fetched 325 MB in 55s (5900 kB/s)
Selecting previously unselected package elasticsearch.
(Reading database ... 61744 files and directories currently installed.)
Preparing to unpack .../elasticsearch_7.17.29_amd64.deb ...
Creating elasticsearch group... OK
Creating elasticsearch user... OK
Unpacking elasticsearch (7.17.29) ...
Setting up elasticsearch (7.17.29) ...
### NOT starting on installation, please execute the following statements to configure elasticsearch s
ervice to start automatically using systemd
  sudo systemctl daemon-reload
  sudo systemctl enable elasticsearch.service
### You can start elasticsearch service by executing
  sudo systemctl start elasticsearch.service
Created elasticsearch keystore in /etc/elasticsearch/elasticsearch.keystore
ishwor@OHH:~/bigdata$ |
```

```
ishwor@OHH:~/bigdata$ sudo nano /etc/elasticsearch/elasticsearch.yml
```

```

GNU nano 6.2 /etc/elasticsearch/elasticsearch.yml
# ===== Elasticsearch Configuration =====
#
# NOTE: Elasticsearch comes with reasonable defaults for most settings.
#       Before you set out to tweak and tune the configuration, make sure you
#       understand what are you trying to accomplish and the consequences.
#
# The primary way of configuring a node is via this file. This template lists
# the most important settings you may want to configure for a production cluster.
#
# Please consult the documentation for further information on configuration options:
# https://www.elastic.co/guide/en/elasticsearch/reference/index.html
#
# ----- Cluster -----
#
# Use a descriptive name for your cluster:
#
#cluster.name: my-application
#
# ----- Node -----
#
# Use a descriptive name for the node:
#
#node.name: node-1
#
# Add custom attributes to the node:
#
#node.attr.rack: r1
#
# ----- Paths -----
#
# Path to directory where to store the data (separate multiple locations by comma):
#
path.data: /var/lib/elasticsearch
#
# Path to log files:
#
path.logs: /var/log/elasticsearch
#
# ----- Memory -----
#
# Lock the memory on startup:
#
#bootstrap.memory_lock: true
#
# Make sure that the heap size is set to about half the memory available
# on the system and that the owner of the process is allowed to use this
# limit.
#
# Elasticsearch performs poorly when the system is swapping the memory.

```

```

ishwor@OHH:~/bigdata$ sudo systemctl start elasticsearch
ishwor@OHH:~/bigdata$ sudo systemctl enable elasticsearch
Synchronizing state of elasticsearch.service with SysV service script with /lib/systemd/systemd-sysv-i
nstall.
Executing: /lib/systemd/systemd-sysv-install enable elasticsearch
Created symlink /etc/systemd/system/multi-user.target.wants/elasticsearch.service → /lib/systemd/syste
m/elasticsearch.service.
ishwor@OHH:~/bigdata$ |

```

```

ishwor@OHH:~/bigdata$ sudo apt install kibana
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  htop libnl-genl-3-200
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  kibana
0 upgraded, 1 newly installed, 0 to remove and 55 not upgraded.
Need to get 296 MB of archives.
After this operation, 753 MB of additional disk space will be used.
Get:1 https://artifacts.elastic.co/packages/7.x/apt stable/main amd64 kibana amd64 7.17.29 [296 MB]
Fetched 296 MB in 49s (6065 kB/s)
Selecting previously unselected package kibana.
(Reading database ... 62843 files and directories currently installed.)
Preparing to unpack .../kibana_7.17.29_amd64.deb ...
Unpacking kibana (7.17.29) ...
Setting up kibana (7.17.29) ...
Creating kibana group... OK
Creating kibana user... OK
Kibana is currently running with legacy OpenSSL providers enabled! For details and instructions on how
to disable see https://www.elastic.co/guide/en/kibana/7.17/production.html#openssl-legacy-provider
Created Kibana keystore in /etc/kibana/kibana.keystore
ishwor@OHH:~/bigdata$ sudo systemctl start kibana

```

```

ishwor@OHH:~/bigdata$ sudo apt update
Hit:1 https://artifacts.elastic.co/packages/7.x/apt stable InRelease
Hit:2 http://archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:4 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:5 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu jammy InRelease
Hit:6 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
55 packages can be upgraded. Run 'apt list --upgradable' to see them.
ishwor@OHH:~/bigdata$ sudo apt-get install nginx
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  htop libnl-genl-3-200
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libnginx-mod-http-geoip2 libnginx-mod-http-image-filter libnginx-mod-http-xslt-filter
  libnginx-mod-mail libnginx-mod-stream libnginx-mod-stream-geoip2 libxslt1.1 nginx-common
  nginx-core
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  libnginx-mod-http-geoip2 libnginx-mod-http-image-filter libnginx-mod-http-xslt-filter
  libnginx-mod-mail libnginx-mod-stream libnginx-mod-stream-geoip2 libxslt1.1 nginx nginx-common
  nginx-core
0 upgraded, 10 newly installed, 0 to remove and 55 not upgraded.
Need to get 863 kB of archives.
After this operation, 2907 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 nginx-common all 1.18.0-6ubuntu14.8 [4
0.1 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libnginx-mod-http-geoip2 amd64 1.18.0-
6ubuntu14.8 [12.1 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libnginx-mod-http-image-filter amd64 1
.18.0-6ubuntu14.8 [15.6 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libxslt1.1 amd64 1.1.34-4ubuntu0.22.04
.5 [165 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libnginx-mod-http-xslt-filter amd64 1.
18.0-6ubuntu14.8 [13.9 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libnginx-mod-mail amd64 1.18.0-6ubuntu

```

```

ishwor@OHH:~/bigdata$ echo "kibanaadmin:`openssl passwd -apr1`" | sudo tee -a
/etc/nginx/htpasswd.users
Password:
Verifying - Password:
kibanaadmin:$apr1$77K9c5bF$vEjEz.xRHYyju.Lrn.Klu1
-bash: /etc/nginx/htpasswd.users: No such file or directory
ishwor@OHH:~/bigdata$ sudo nano /etc/nginx/sites-available/elk
ishwor@OHH:~/bigdata$ sudo ln -s /etc/nginx/sites-available/elk /etc/nginx/sites-enabled/elk
ishwor@OHH:~/bigdata$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
ishwor@OHH:~/bigdata$ sudo systemctl reload nginx
ishwor@OHH:~/bigdata$

```

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localhost:5601/status

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Kibana status is Green

OHH

1.90 GB
 Heap total

181.91 MB
 Heap used

0.00
 Requests per second

0.13, 0.20, 0.12
 Load
1m; 5m; 15m
 Load interval

10.94 ms
 Delay avg
50: 11.02 ms; 95: 11.13 ms; 99: 11.20 ms
 Percentiles

0.00 ms
 Response time avg
0.00 ms
 Response time max

Plugin status

BUILD 47815 COMMIT f6e2f2e44cc0a6a11435f1b6350f735e23bef4b4

ID	Status
core:elasticsearch	Elasticsearch is available
core:savedObjects	SavedObjects service has completed migrations and is available
plugin:licensing	License fetched
plugin:banners	All dependencies are available
plugin:features	All dependencies are available
plugin:globalSearch	All dependencies are available
plugin:globalSearchProviders	All dependencies are available
plugin:uiActionsEnhanced	All dependencies are available

```

ishwor@OHH:~/bigdata$ sudo apt install logstash
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  htop libnl-genl-3-200
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  logstash
0 upgraded, 1 newly installed, 0 to remove and 55 not upgraded.
Need to get 375 MB of archives.
After this operation, 632 MB of additional disk space will be used.
Get:1 https://artifacts.elastic.co/packages/7.x/apt stable/main amd64 logstash amd64 1:7.17.29-1 [375 MB]
Fetched 375 MB in 1min 26s (4372 kB/s)
Selecting previously unselected package logstash.
(Reading database ... 109367 files and directories currently installed.)
Preparing to unpack .../logstash_1%3a7.17.29-1_amd64.deb ...
Unpacking logstash (1:7.17.29-1) ...
Setting up logstash (1:7.17.29-1) ...
Using bundled JDK: /usr/share/logstash/jdk
Using provided startup.options file: /etc/logstash/startup.options
OpenJDK 64-Bit Server VM warning: Option UseConcMarkSweepGC was deprecated in version 9.0 and will likely be removed in a future release.
/usr/share/logstash/vendor/bundle/jruby/2.5.0/gems/pleaserun-0.0.32/lib/pleaserun/platform/base.rb:112: warning: constant ::Fixnum is deprecated
Successfully created system startup script for Logstash
ishwor@OHH:~/bigdata$ sudo nano /etc/logstash/conf.d/02-beats-input.conf
ishwor@OHH:~/bigdata$

```

```

ishwor@OHH:~/bigdata$ sudo nano /etc/logstash/conf.d/30-elasticsearch-output.conf
ishwor@OHH:~/bigdata$ sudo -u logstash /usr/share/logstash/bin/logstash --path.settings /etc/logstash -t
Using bundled JDK: /usr/share/logstash/jdk
OpenJDK 64-Bit Server VM warning: Option UseConcMarkSweepGC was deprecated in version 9.0 and will likely be removed in a future release.
[FATAL] 2026-04-23 10:46:50.322 [main] Logstash - Logstash stopped processing because of an error: (ENOENT) No such file or directory - /home/ishwor/bigdata
org.jruby.exceptions.SystemCallError: (ENOENT) No such file or directory - /home/ishwor/bigdata
    at org.jruby.RubyDir.chdir(org/jruby/RubyDir.java:424) ~[jruby-complete-9.2.20.1.jar:?]
    at usr.share.logstash.vendor.bundle.jruby.$2_dot_5_dot_0.gems.bundler_minus_2_dot_3_dot_18.lib.bundler.shared_helpers.chdir(/usr/share/logstash/vendor/bundle/jruby/2.5.0/gems/bundler-2.3.18/lib/bundler/shared_helpers.rb:52) ~[?:?]
    at uri_3a_classloader_3a_META_minus_INF.jruby_dot_home.lib.ruby.stdlib.monitor.mon_synchronize(uri:classloader:/META-INF/jruby.home/lib/ruby/stdlib/monitor.rb:237) ~[?:?]
    at org.jruby.RubyThread.handle_interrupt(org/jruby/RubyThread.java:759) ~[jruby-complete-9.2.20.1.jar:?]
    at uri_3a_classloader_3a_META_minus_INF.jruby_dot_home.lib.ruby.stdlib.monitor.mon_synchronize(uri:classloader:/META-INF/jruby.home/lib/ruby/stdlib/monitor.rb:236) ~[?:?]
    at org.jruby.RubyThread.handle_interrupt(org/jruby/RubyThread.java:759) ~[

```



```
ishwor@OHH:~/bigdata$ sudo systemctl start logstash
ishwor@OHH:~/bigdata$ sudo systemctl enable logstash
Created symlink /etc/systemd/system/multi-user.target.wants/logstash.service → /etc/systemd/system/logstash.service.
ishwor@OHH:~/bigdata$
```

```
ishwor@OHH:~/bigdata$ sudo apt install filebeat
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  htop libnl-genl-3-200
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  filebeat
0 upgraded, 1 newly installed, 0 to remove and 55 not upgraded.
Need to get 37.5 MB of archives.
After this operation, 137 MB of additional disk space will be used.
Get:1 https://artifacts.elastic.co/packages/7.x/apt stable/main amd64 filebeat amd64 7.17.29 [37.5 MB]
Fetched 37.5 MB in 4s (9906 kB/s)
Selecting previously unselected package filebeat.
(Reading database ... 124793 files and directories currently installed.)
Preparing to unpack .../filebeat_7.17.29_amd64.deb ...
Unpacking filebeat (7.17.29) ...
Setting up filebeat (7.17.29) ...
ishwor@OHH:~/bigdata$ sudo nano /etc/filebeat/filebeat.yml
ishwor@OHH:~/bigdata$ |
```



```
ishwor@OHH:~/bigdata$ sudo filebeat modules enable system
```

```
Enabled system
```

```
ishwor@OHH:~/bigdata$ sudo filebeat modules list
```

```
Enabled:
```

```
system
```

```
Disabled:
```

```
activemq
```

```
apache
```

```
auditd
```

```
aws
```

```
awsfargate
```

```
azure
```

```
barracuda
```

```
bluecoat
```

```
cef
```

```
checkpoint
```

```
cisco
```

```
coredns
```

```
crowdstrike
```

```
cyberark
```

```
cyberarkpas
```

```
cylance
```

```
elasticsearch
```

```
envoyproxy
```

```
f5
```

```
fortinet
```

```
gcp
```

```
google_workspace
```

```
googlecloud
```

```
gsuite
```

```
haproxy
```

```
ibmmq
```

```
icinga
```

```
iis
```

```
imperva
```

```
infoblox
```

```
iptables
```

```
juniper
```

```
logstash
```

```
ishwor@OHH:~/bigdata$ sudo filebeat setup -E output.logstash.enabled=false -E \
output.elasticsearch.hosts=['localhost:9200'] -E \
setup.kibana.host=localhost:5601
Overwriting ILM policy is disabled. Set `setup.ilm.overwrite: true` for enabling.
```

```
Index setup finished.
```

```
Loading dashboards (Kibana must be running and reachable)
```

```
Loaded dashboards
```

```
Setting up ML using setup --machine-learning is going to be removed in 8.0.0. Please use the ML app instead.
```

```
See more: https://www.elastic.co/guide/en/machine-learning/current/index.html
```

```
It is not possible to load ML jobs into an Elasticsearch 8.0.0 or newer using the Beat.
```

```
Loaded machine learning job configurations
```

```
Loaded Ingest pipelines
```

```
ishwor@OHH:~/bigdata$
```

```
ishwor@OHH:~/bigdata$ sudo systemctl start filebeat
ishwor@OHH:~/bigdata$ sudo systemctl enable filebeat
Synchronizing state of filebeat.service with SysV service script with /lib/systemd
/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable filebeat
Created symlink /etc/systemd/system/multi-user.target.wants/filebeat.service → /li
b/systemd/system/filebeat.service.
ishwor@OHH:~/bigdata$ curl -XGET 'http://localhost:9200/filebeat-*/_search?pretty'
{
  "took" : 3,
  "timed_out" : false,
  "_shards" : {
    "total" : 1,
    "successful" : 1,
    "skipped" : 0,
    "failed" : 0
  },
  "hits" : {
    "total" : {
      "value" : 0,
      "relation" : "eq"
    },
    "max_score" : null,
    "hits" : [ ]
  }
}
```

workshop - Google Drive xInstalling ELK.docx - Google xElastic xHome - ElasticFPS N/A | GPU 0% | CPU 14% | RAM N/A

localhost:5601/app/home#/

2026 - 6CS012 - Al...Colab Notebooks ~..._6CS030 BigData ~...50 Beautiful Websit...

elasticSearch Elastic

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Welcome home



Enterprise Search

Create search experiences with a refined set of APIs and tools.



Observability

Consolidate your logs, metrics, application traces, and system availability with purpose-built UIs.



Security

Prevent, collect, detect, and respond to threats for unified protection across your infrastructure.



Analytics

Explore, visualize, and analyze your data using a powerful suite of analytical tools and applications.

Get started by adding integrations

To start working with your data, use one of our many ingest options. Collect data from an app or service, or upload a file. If you're not ready to use your own data, add a sample data set.

+ Add integrations

Try sample data

Upload a file



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Installing ELK.docx - Google

Elastic

Home - Elastic

FPS N/A

GPU 0%

CPU 10%

RAM N/A

localhost:5601/status

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Kibana status is Green

OHH

1.90 GB

Heap total

173.82 MB

Heap used

0.20

Requests per second

0.33, 0.37, 0.26

Load

10.82 ms

Delay avg

366.00 ms

Response time avg

1m; 5m; 15m

Load interval

50: 10.96 ms; 95: 11.12 ms; 99: 11.18 ms

Percentiles

366.00 ms

Response time max

Plugin status

BUILD 47815 COMMIT f6e2f2e44cc0a6a11435f1b6350f735e23bef4b4

ID	Status
core:elasticsearch	Elasticsearch is available
core:savedObjects	SavedObjects service has completed migrations and is available
plugin:licensing	License fetched
plugin:banners	All dependencies are available
plugin:features	All dependencies are available

Workshop 9

1. What is ELK Stack? Explain its architecture.

Answer:

The **ELK Stack** is a combination of three open-source tools used for **centralized logging, data analysis, and visualization**. It consists of:

- **Elasticsearch** – A search and analytics engine
- **Logstash** – A data processing pipeline
- **Kibana** – A data visualization tool

Architecture of ELK Stack

The ELK Stack follows a pipeline architecture:

Data Sources → Logstash → Elasticsearch → Kibana → User

Explanation:

- **Data Sources:**
Applications, servers, and systems generate logs and events.
- **Logstash:**
Collects data from multiple sources, processes it, and converts it into a structured format.
- **Elasticsearch:**
Stores the processed data and allows fast searching and analysis using indexing.
- **Kibana:**
Visualizes the data using dashboards, charts, and graphs for easy understanding.

Conclusion:

The ELK Stack enables real-time monitoring, analysis, and visualization of large-scale data.

2. What is a Shard in Elasticsearch?

Answer:

A **Shard** in Elasticsearch is a **small partition of an index** that allows data to be distributed across multiple nodes.

Types of Shards:

1. **Primary Shard:**
Stores the original data.
2. **Replica Shard:**
A copy of the primary shard used for backup and improved performance.

Importance of Shards:

- Improves **scalability** by splitting large data
- Enables **parallel processing**
- Provides **fault tolerance**
- Enhances **search performance**

Example:

If a large dataset is divided into multiple shards, each shard can be stored on different servers, making the system faster and more reliable.

3. What are the 3 major components of Logstash?

Answer:

Logstash processes data using three main components:

a. Input

- Collects data from various sources such as files, databases, and APIs.

b. Filter

- Processes and transforms the data.
- Performs operations like parsing, cleaning, and formatting.

c. Output

- Sends the processed data to destinations like Elasticsearch or other storage systems.

Flow:

Input → Filter → Output

4. Case Study: Usage of ELK Stack by Netflix

Answer:

One company that effectively uses the ELK Stack is Netflix.

Problem:

Netflix handles massive amounts of data generated from:

- User activity
- Streaming logs
- System performance

Managing and analyzing this data in real-time is challenging.

Solution Using ELK Stack:

- **Logstash:**
Collects and processes logs from multiple microservices.
- **Elasticsearch:**
Stores and indexes large volumes of log data for fast search and analysis.
- **Kibana:**
Provides dashboards for monitoring system health and user behavior.

Use Cases:

- Real-time monitoring of streaming performance
- Detection of system errors and failures
- User behavior analysis
- Security monitoring

Benefits:

- Scalable data processing
- Faster debugging and issue resolution
- Improved user experience

5. Explore Fuzzy Search in ELK

Answer:

Fuzzy Search in ELK allows users to search for data even when the search term contains **spelling errors or variations**.

Concept:

It is based on **edit distance (Levenshtein Distance)**, which measures how many changes are needed to transform one word into another.

Example:

- Search term: *netflx*
- Result: *Netflix*

Elasticsearch Query Example:

```
{
  "query": {
    "fuzzy": {
      "name": {
        "value": "netflx",
        "fuzziness": "AUTO"
      }
    }
  }
}
```

Applications:

- Handling user typing errors
- Improving search accuracy
- Log searching with approximate matches

Advantages:

- Better user experience
- Flexible search results
- Increased accuracy in data retrieval

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localhost:5601/app/dev_tools#/console

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```
1 GET _search
2 {
3   "query": {
4     "match_all": {}
5   }
6 }
```

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localhost:9200

Pretty print

```
{
  "name" : "OHH",
  "cluster_name" : "elasticsearch",
  "cluster_uuid" : "hTKVswzpITIKX9PUwiOKBZw",
  "version" : {
    "number" : "7.17.29",
    "build_flavor" : "default",
    "build_type" : "deb",
    "build_hash" : "580aff1a0064ce4c93293aaab6fcc55e22c10d1c",
    "build_date" : "2025-06-19T01:37:57.847711500Z",
    "build_snapshot" : false,
    "lucene_version" : "8.11.3",
    "minimum_wire_compatibility_version" : "6.8.0",
    "minimum_index_compatibility_version" : "6.0.0-beta1"
  },
  "tagline" : "You Know, for Search"
}
```

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1 GET /_cat/health?v

200 - OK 53 ms

```
1 #! Elasticsearch built-in security features are not enabled.
  Without authentication, your cluster could be accessible to
  anyone. See https://www.elastic.co/guide/en/elasticsearch
  /reference/7.17/security-minimal-setup.html to enable security.
2 epoch      timestamp cluster      status node.total node.data
  shards pri relo init unassign pending_tasks max_task_wait_time
  active_shards_percent
3 1776921646 05:20:46 elasticsearch yellow      1          1
   9 9 0 0 1 0
   90.0%
```

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History Settings Help

1 PUT /resourcename

200 - OK 293 ms

```
1 #! Elasticsearch built-in security features are not enabled.
  Without authentication, your cluster could be accessible to
  anyone. See https://www.elastic.co/guide/en/elasticsearch
  /reference/7.17/security-minimal-setup.html to enable security.
2 {
3   "acknowledged" : true,
4   "shards_acknowledged" : true,
5   "index" : "resourcename"
6 }
7
```

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1GET /_cat/indices

200 - OK27 ms

1#! Elasticsearch built-in security features are not enabled. Without authentication, your cluster could be accessible to anyone. See <https://www.elastic.co/guide/en/elasticsearch/reference/7.17/security-minimal-setup.html> to enable security.

2green open .geoip_databasesp9F2CAi1R9e

3green open .kibana_task_manager_7.17.29_001

4green open .kibana_7.17.29_001

5green open .apm-custom-linkSHbFSi1

6green open .apm-agent-configurationTg2xL4SUQbSGcB8

7yellow open filebeat-7.17.29-2026.04.23-000001 cYjqACEiTG

8yellow open resourcename

9k95Bw4RzTF6QvnbjPOscYw 1 1 0 0 227b 227b

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HistorySettingsHelp

1PUT /students/colname/1

2{

3 "name": "John Doe"

4}

5

1#! Elasticsearch built-in security features are not enabled. Without authentication, your cluster could be accessible to anyone. See <https://www.elastic.co/guide/en/elasticsearch/reference/7.17/security-minimal-setup.html> to enable security.

2#! [types removal] Specifying types in document index requests is deprecated, use the typeless endpoints instead (/_{index}/_doc/{id}, /_{index}/_doc, or /_{index}/_create/{id}).

3{

4 "_index" : "students",

5 "_type" : "colname",

6 "_id" : "1",

7 "_version" : 4,

8 "result" : "updated",

9 "_shards" : {

10 "total" : 2,

11 "successful" : 1,

12 "failed" : 0

13 },

14 "_seq_no" : 3,

15 "_primary_term" : 1

16}

17

200 - OK15 ms

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HistorySettingsHelp

1PUT /resourcename/colname/2

2{

3 "name": "Your Name",

4 "class": "Your Class Group",

5 "email": "your.email@example.com"

6 }

7

201 - Created96 ms

1#! Elasticsearch built-in security features are not enabled. Without authentication, your cluster could be accessible to anyone. See <https://www.elastic.co/guide/en/elasticsearch/reference/7.17/security-minimal-setup.html> to enable security.

2#! [types removal] Specifying types in document index requests is deprecated, use the typeless endpoints instead (/index/_doc/{id}, /index/_doc, or /index/_create/{id}).

3{

4 "_index": "resourcename",

5 "_type": "colname",

6 "_id": "2",

7 "_version": 1,

8 "result": "created",

9 "shards": {

10 "total": 2,

11 "successful": 1,

12 "failed": 0

13 },

14 "_seq_no": 0,

15 "_primary_term": 1

16 }

17

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HistorySettingsHelp

1 GET /resourcename/_search2

200 - OK235 ms

1 #! Elasticsearch built-in security features are not enabled. Without authentication, your cluster could be accessible to anyone. See <https://www.elastic.co/guide/en/elasticsearch/reference/7.17/security-minimal-setup.html> to enable security.

2 {

3 "took" : 240,

4 "timed_out" : false,

5 "_shards" : {

6 "total" : 1,

7 "successful" : 1,

8 "skipped" : 0,

9 "failed" : 0

10 },

11 "hits" : {

12 "total" : {

13 "value" : 1,

14 "relation" : "eq"

15 },

16 "max_score" : 1.0,

17 "hits" : [

18 {

19 "_index" : "resourcename",

20 "_type" : "colname",

21 "_id" : "2",

22 "_score" : 1.0,

23 "_source" : {

24 "name" : "Your Name",

25 "class" : "Your Class Group",

26 "email" : "your.email@example.com"

27 }

28 }

29]

30 }

31 }

32 }

Exercise to do

1. **Write an API query call to fetch the second data record (that was your personal details).**

Answer:

GET /users?limit=1&offset=1

Explanation:

- limit=1 → retrieves only one record
- offset=1 → skips the first record

Therefore, this query returns the **second data record**.

2. **Write a query to remove the data record of John Doe from the resource.**

Answer:

DELETE /users/2

Explanation:

- DELETE is the HTTP method used to remove data
- /users/2 refers to the user with ID = 2 (assumed to be John Doe)

Alternative (if API supports query parameters):

DELETE /users?name=John%20Doe

Explanation:

- Deletes the record where the name is **John Doe**